

Unit 14, Lesson 6: Solving One-Step Multiplication and Division Equations in One Variable Using Models



Benchmark

MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.



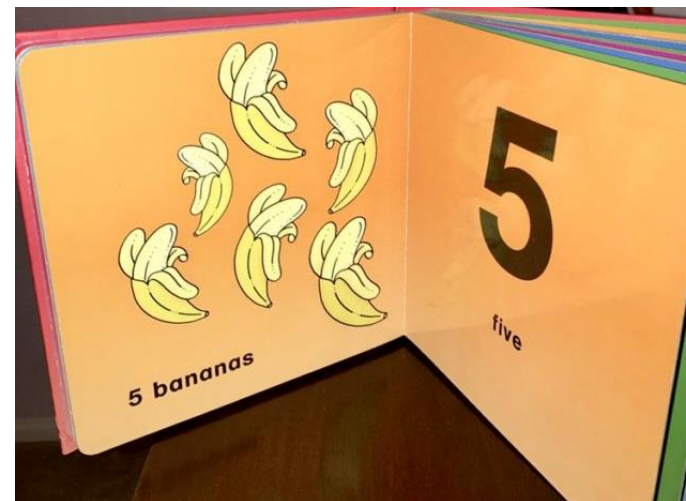
Learning Targets

- ☐ I can solve one-step multiplication and division equations using manipulatives.
- ☐ I can solve one-step multiplication and division equations using diagrams.



14.6.1 Warm-Up: Math Fail

This children's book has published a "math fail."



1. Explain the error shown in this image.



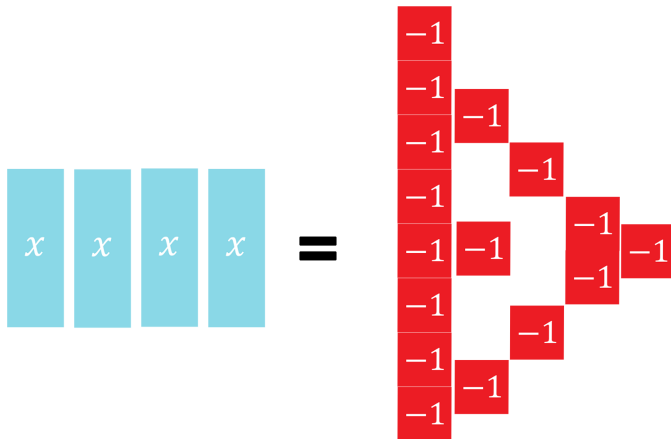
14.6.2 Exploration: Exploring Relationships

Recall that when solving an equation, the mathematical sentence must remain balanced so that the two expressions remain equivalent.

1. A model of the equation $4x = -16$ is shown using algebra tiles.

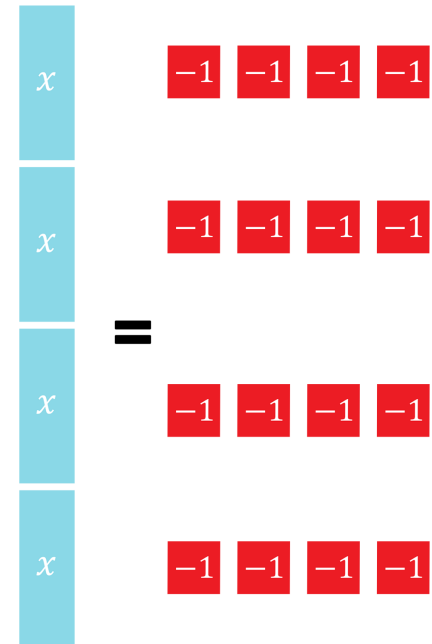


The  tile represents the variable, x , and the  represents the constant term where each tile is 1 unit.

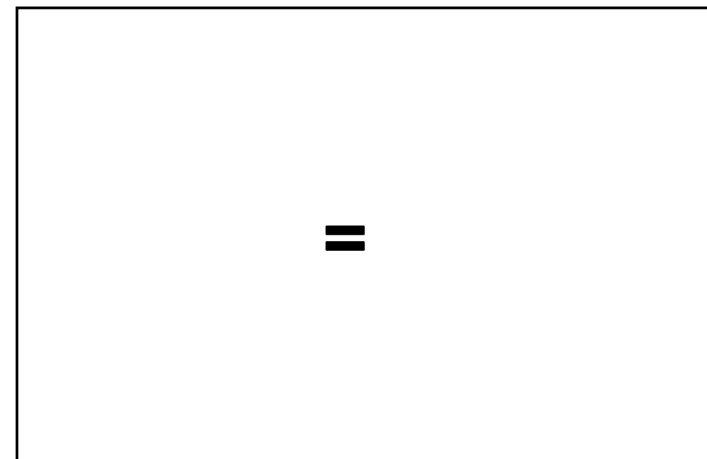


- a. What operation is being applied to the 4 and the x in the equation $4x = -16$?

- b. The algebra tiles are arranged on the equation mat into equal-sized groups on both sides of the equation. How many groups were created?

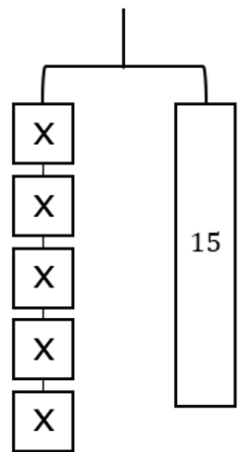


- c. Use algebra tiles to sketch the solution of the equation.



- d. What is the value of x ?

2. The hanger diagram represents the equation $5x = 15$.



- a. Complete the statements.

The left side of the hanger is broken into

- ☐ 3
- ☐ 5
- ☐ 15

equal parts. If 15 is broken into the same number of equal parts, each part will have the same weight as

one x block. Splitting each side of the hanger into

- ☐ 3
- ☐ 5
- ☐ 15

the same as ☐ dividing ☐ multiplying each

side of the equation by

- ☐ 3.
- ☐ 5.
- ☐ 15.

- b. Sketch splitting the hanger diagram into the equal parts as described in Part A.

- c. What is the value of x ?

- d. Compare your diagram with your partner. Discuss the similarities and differences.

- e. Provide positive feedback to your partner in their book on their sketch and solution.

Partner Feedback	
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3. Inverse operations can be helpful in solving equations.

- a. Draw an arrow to match each operation on the left with its inverse operation on the right.

division

division

addition

addition

subtraction

subtraction

multiplication

multiplication

- b. Explain what it means for operations to be inverses.

- c. Which inverse operations were used to solve the equations in this exploration?



14.6.3 Guided Instruction: Using Models to Solve Equations

Algebra tiles can also be used to model division equations.

1. The equation $\frac{x}{3} = 4$ can be rewritten as $x \div 3 = 4$.

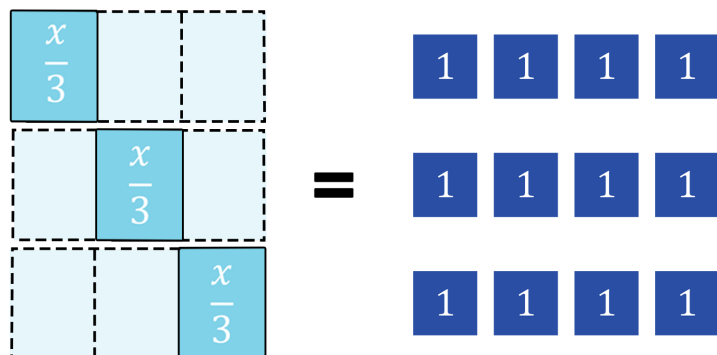
To represent division, show that x tile is being divided into equal parts. Sketch lines on the x tile to represent the fraction $x \div 3$.

This means when x is divided into 3 parts, each part is equal to 4. The model shown illustrates that $\frac{x}{3} = 4$.

$$\frac{x}{3} = 4$$

- a. If each $\frac{1}{3}$ of x is equal to 4, discuss with your partner what the value of x is. As part of your discussion, include which operation can be used to determine the value of x . Summarize your discussion.

- b. The image models the relationship discussed that can be used to determine the value of x .



Explain how this representation can be used to determine the value of x .

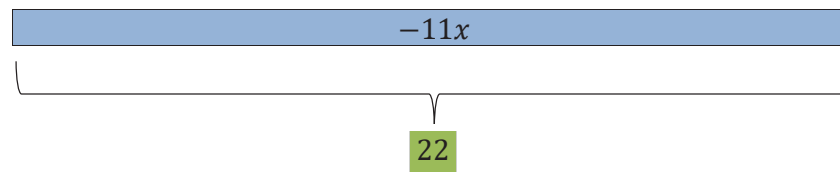
- c. What is the value of x ?

- d. Complete the statements by writing operations on each blank.

The original equation involved the operation _____ . Its inverse operation, _____ , was used to solve the equation.

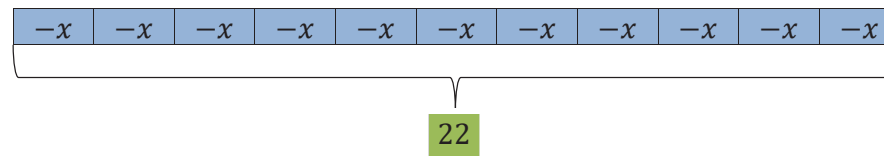
Tape diagrams, also known as bar models, can be used to model and solve multiplication and division equations.

2. The tape diagram models the equation $22 = -11x$.



The solution to the equation $22 = -11x$ can be modeled in the following steps. Work with a different partner to explain what is occurring in each step.

Step 1



- a. How were the integers in the equation used to determine the negative value of x ?

Explanation of Step 1

Step 2

$$\frac{-x}{2}$$

Explanation of Step 2

Step 3

$$\frac{x}{-2}$$

Explanation of Step 3

- b. Discuss with your partner how the operation used in the equation relates to the operation used in the solution. Summarize your discussion.



14.6.4 Collaboration: Matching Equations and Solutions

1. Match each equation to the correct solution. Record the equation with its solution in the table. Some solutions may be used more than once or not at all. Include an explanation for the match.

A. $x = 2$	E. $x = 3$
B. $x = -20$	F. $x = 10$
C. $x = -10$	G. $x = -8$
D. $x = 9$	H. $x = -3$

1. $-3x = -6$

2. $-\frac{x}{2} = 4$

3. $\frac{x}{3} = 3$

4. $-\frac{x}{2} = 5$

5. $3x = 9$

6. $\frac{x}{10} = -2$

7. $4x = 8$

Equation	Solution	Explanation for Match



14.6.5 Wrap-Up: Ticket Out the Door

- Throughout this unit, you have been shown how to solve equations using tape diagrams, hanger diagrams, number lines, manipulatives, and inverse operations. Use one of these methods to model and solve the equation $6h = -12$.